Obesity and Its Factors

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Link to Video:
https://youtu.be/t3BWdGCif4M?si=699jRCZKTVSdpkKp

Project Topic

We investigated the drivers of the U.S. obesity epidemic by analyzing the relationship between socioeconomic status (Census data) and food environment (USDA data), specifically focusing on income inequality and access to high and low quality food.

Project Objectives

1. Quantify the correlation between median household income and obesity rates
2. Look into "food deserts" and their impact on obesity rates (high ratio of fast food to grocery stores)
3. Use K-Means clustering to identify distinct risk profiles for U.S. counties

Data Collection / Cleaning

Downloaded Dataset Source:
https://www.ers.usda.gov/data-products/food-environment-atlas/data-access-and-documentation-downloads
Web Collection #1 Source:
https://en.wikipedia.org/wiki/Obesity_in_the_United_States
Web Collection #2 Source:
https://api.census.gov/data/2022/acs/acs5
Modules:
Pandas, Matplotlib, Sci-Kit Learn (LinearRegression, KMeans), Beautiful Soup

Insight 1

The Income Impact:
Data showed a moderate negative correlation (-0.32). The wealthier communities become, the lower obesity rates they have, likely due to better access to healthcare and fitness resources.

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Correlation: -0.3228

Avg Obesity by Income Level:
Income_Group
Low          33.191221
Mid-Low      32.308651
Mid-High     31.359236
High         30.640456
```

Insight 2

The "Food Desert" Issue:
Counties with high ratios of fast food to grocery stores actually had a slightly lower obesity rate than low ratio areas, suggesting against the theory that easier access to fast food drives obesity.

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Obesity in High Fast Food Areas: 32.39%
Obesity in Low Fast Food Areas: 32.61%
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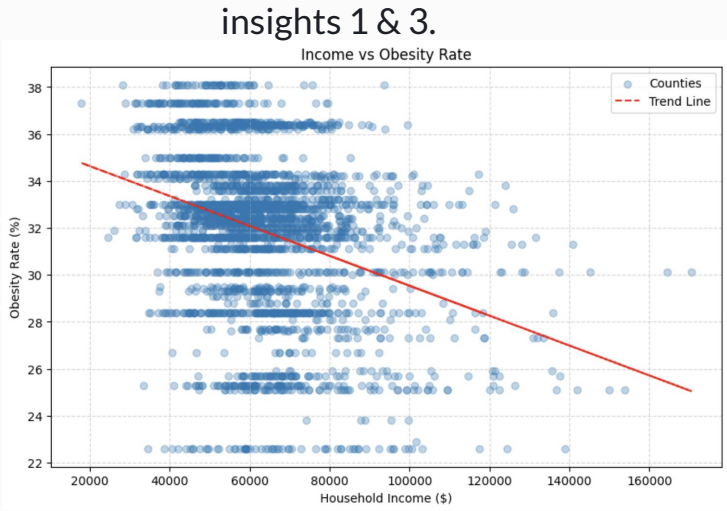
Insight 3

Demographic Clustering:
Using Sci-Kit Learn K-Means, we identified three distinct clusters. The "Low Income" cluster averaged 33.6% obesity, while the "High Income" cluster averaged 30.5%, statistically supporting the idea that economic status is a genuine risk factor for developing obesity.

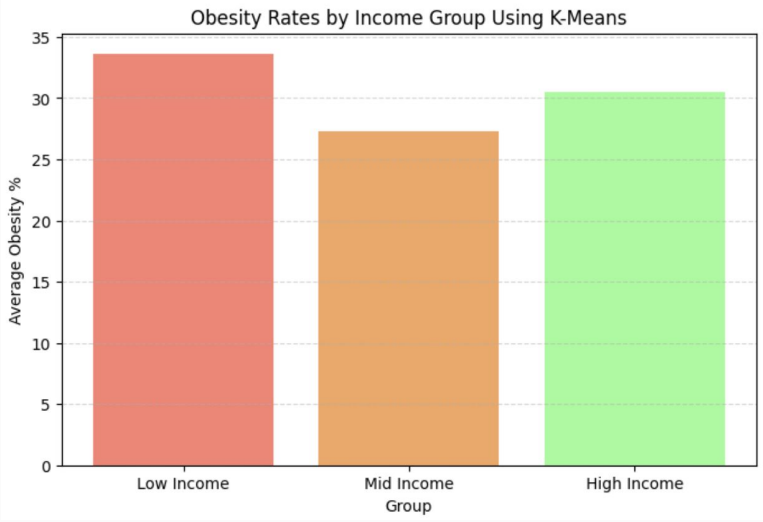
Visual 1

(Scatter Plot: Income vs. Obesity) Shows negative correlation through red trend line, supporting insights 1 & 3.



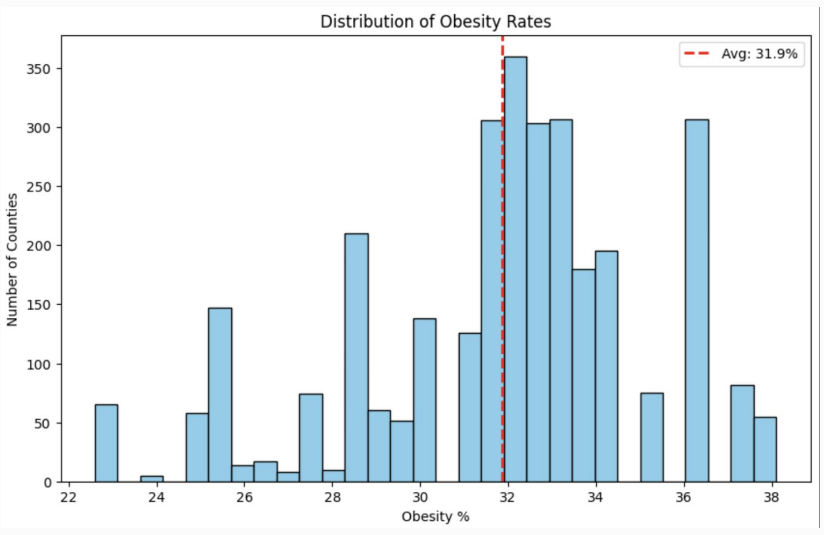
Visual 2

(Bar Chart: K-Means Clusters) Shows Income Tiers vs. Obesity, supporting insights 1 & 3.



Visual 3

(Histogram) Shows right skewed distribution of obesity rates, suggesting more counties face high rates of obesity than low rates.

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Group Averages:
Labeled Group Averages:
                Group_Label  Median_Household_Income  Adult_Obesity_Percent
0  High Income / Low Obesity              92112.378661              30.532501
1  Low Income / High Obesity              62998.988871              27.275199
2               Middle Class              56646.981336              33.611328
```

	Median_Household_Income	Adult_Obesity_Percent	Group	Group_Label
0	68315.0	36.300000	1	Middle Class
1	71039.0	36.300000	1	Middle Class
2	39712.0	36.300000	1	Middle Class
3	50669.0	36.300000	1	Middle Class
4	57440.0	36.300000	1	Middle Class
...	...	...	...	...
3138	83119.0	31.863526	2	High Income / Low Obesity
3139	87971.0	31.863526	2	High Income / Low Obesity
3140	83617.0	31.863526	2	High Income / Low Obesity
3141	80330.0	31.863526	2	High Income / Low Obesity
3142	118930.0	31.863526	2	High Income / Low Obesity

3143 rows × 4 columns

Conclusion

Obesity and income are inversely related. From our insights, we can see that as wealth increases, obesity decreases. Surprisingly, physical food environment was a weaker predictor than socioeconomic status, suggesting that the economic ability to purchase healthy food matters more than geographic proximity.